

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. FEATURE SUMMARY ComicCraft is built with an array of robust functionalities that abstract the complex AI orchestration process away from the user, delivering a seamless, automated digital publishing experience.

3. DETAILED FEATURE BREAKDOWN

- **Feature 1: Customizable Prompt Interface**
 - *Functionality:* A dynamic HTML/CSS frontend form that captures tailored narrative inputs, character configurations, and artistic style preferences (e.g., Anime, Realistic, Sketch) via dropdown menus.
- **Feature 2: AI-Powered Narrative Structuring**
 - *Functionality:* Utilizes Google Gemini 1.5 Flash to automatically dissect a basic user prompt into a highly structured 5-panel JSON array, establishing a logical scene-by-scene roadmap.
- **Feature 3: Creative Script Generation**
 - *Functionality:* Deploys Google Gemini 1.5 Pro to expand the structural outline, generating engaging character dialogues, narrations, and sound effects for each panel.
- **Feature 4: Automated Illustration Engine**
 - *Functionality:* Integrates with the Hugging Face Stable Diffusion API to autonomously render high-quality, genre-specific panel artwork based on the AI-generated scene descriptions.
- **Feature 5: Fail-Safe Execution Mechanisms**
 - *Functionality:* Incorporates deep exception handling that detects network timeouts, IP blocks, or API rate limits, automatically defaulting to Python PIL-generated placeholder graphics to prevent system crashes.
- **Feature 6: Interactive Web Rendering**

- *Functionality:* Leverages the Jinja2 templating engine to iterate through backend arrays, dynamically displaying the generated comic panels, titles, and text sequentially on the user's browser.
- **Feature 7: Programmatic PDF Export**
 - *Functionality:* Embeds the FPDF module to dynamically calculate dimensional layouts and bind the visual and textual elements into a multi-page, offline-ready PDF document for the end-user.